More web CAD + FEM pairs

Collected 12 October 2026 | Original published assets | No local geometry generation or FEM solves

New source	Downloaded and verified
AgentFEM bracket elasticity	176 STEP + mesh + displacement/stress pairs; 128 train / 16 validation / 32 test. One parametric family.
CADWorld FEM	3 native FreeCAD projects + saved CalculiX results + CSVs + task/evaluator JSON. Two geometry candidates; one basic specimen.

179 new CAD/result pairs. These are candidates, not 179 distinct complex engineering task types. They extend the collection; earlier aircraft, implant, crash, forming, valve and WAAM sources remain available.

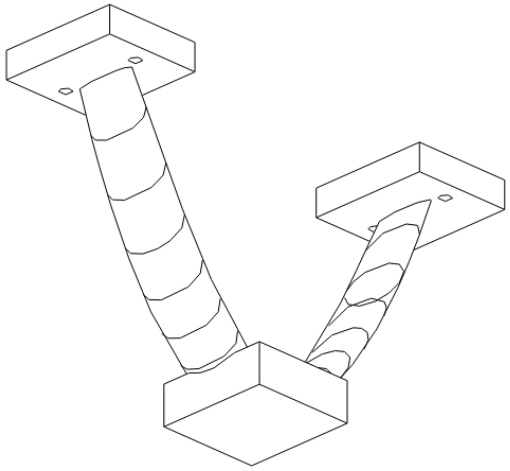
Zero new certified benchmark admissions. Bracket STEP unit metadata conflicts with field coordinates. CADWorld references include very high elastic stress/deformation. Preserve the published data and resolve these issues before judging engineering accuracy.

The supplied-CAD track gains original native geometry and numerical outputs. No new original dimensioned drawing was verified. Inspection renders and task screenshots are labeled accordingly; they do not complete the drawing-to-CAD track.

Checks include publisher SHA-256 hashes where provided, ZIP CRC, native solid validity, finite numerical fields, source load correspondence and cross-checks of reference values. No frontier model trial or independent solver replay was performed.

Sources:
AgentFEM-Bracket-Elasticity (CC BY 4.0, Haoming Luo).
CADWorld repository; publisher dataset declares Apache-2.0. Embedded material-card licenses remain intact.

Bracket elasticity: CAD and FEM references



Inspection render of downloaded fresh_test_000 STEP solid. Not an original dimensioned drawing. Geometry has two feet and curved support webs; bonded feet and uniform downward seat traction replace bolt contact.

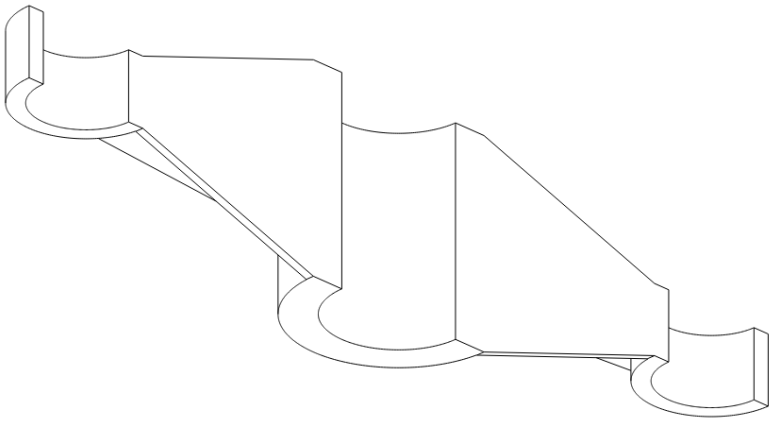
Published result at 10,000 N	Value
Mean seat displacement	0.098354836386 mm
Maximum displacement	0.175628088556 mm
Sampled peak von Mises stress	78.539213022749 MPa
Sampled 95th percentile stress	42.922439729845 MPa

All 176 arrays are finite, fixed-plane displacements are zero, and summary values agree with the original NPZ arrays. Material: E = 70 GPa, nu = 0.33, density = 2700 kg/m3. Feet: z = 0; load seat: z = 0.189 m. Raw references retain their 1000 or 10000 N load.

UNIT BLOCKER: STEP declares millimetres, but its numeric extents are 0.306 x 0.080 x 0.189 and match FEM metres without conversion. That is a factor-1000 scale inconsistency. Source files are retained unchanged; no corrected CAD is substituted.

Publisher states a nominal 9 mm mesh, no industrial convergence qualification, and no plasticity/fracture/contact validation. Sharp-junction stress is a raw sampled output, not a validated strength score.

Multi-feature mechanical part under end load



Inspection render of original native FreeCAD Body BREP. Multi-feature geometry; linear statics. This is not an original dimensioned drawing.

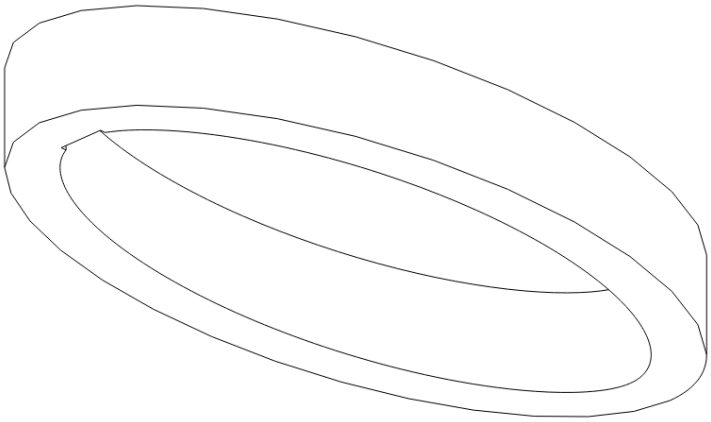
Original native project / CSV	Verified content
CAD	1 solid; 38 faces; valid BREP; editable FCStd retained
Published load	200 N
Result rows	100279
Maximum displacement	31.145 mm
Peak von Mises stress	705.453 MPa

Saved FCStd includes displacement vectors, von Mises arrays, mesh and VTU results. All CSV values are finite. CSV displacement in m agrees with native mm; CSV stress in Pa agrees with native MPa. Source evaluator extrema match the CSV with its stated 5% tolerance.

Publisher task image identifies the fixed and loaded ends. 200 N Y-axis load, AlMg3F24, Gmsh mesh max 2 mm/min 0.5 mm, CalculiX. Large displacement and 705 MPa stress require checking elastic validity and strength interpretation before benchmark certification.

Original task/evaluator JSON, precondition project, solved project and full numerical CSV are included. No source evaluator or solver was executed locally. Independent replay, mesh convergence and physical-validity review remain pending.

Elliptical chain-link tension



Inspection render of original native FreeCAD Body BREP. Curved link geometry; linear statics, no link-link contact verified. This is not an original dimensioned drawing.

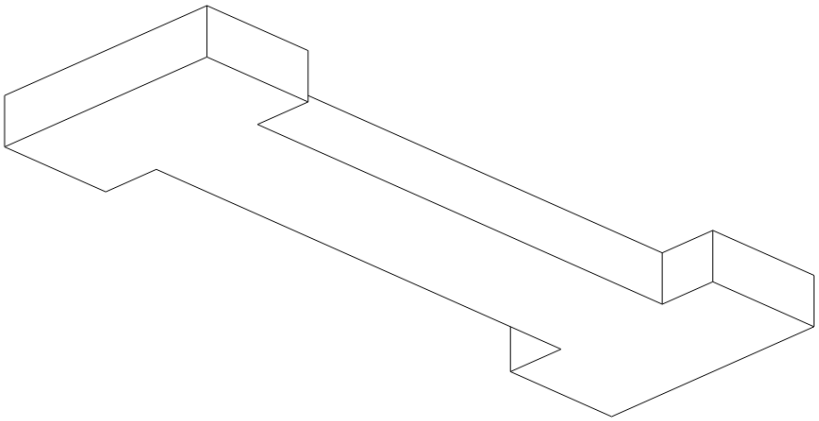
Original native project / CSV	Verified content
CAD	1 solid; 8 faces; valid BREP; editable FCStd retained
Published load	20000 N
Result rows	1349
Maximum displacement	6.96107 mm
Peak von Mises stress	76610.3 MPa

Saved FCStd includes displacement vectors, von Mises arrays, mesh and VTU results. All CSV values are finite. CSV displacement in m agrees with native mm; CSV stress in Pa agrees with native MPa. Source evaluator extrema match the CSV with its stated 5% tolerance.

20 kN tension task, Iron-Generic, Gmsh mesh max 1 mm/min 0.5 mm, CalculiX and displacement WarpVector. Peak stress is about 76.6 GPa: this is a saved software-workflow reference, not a credible elastic strength label. No link-link contact solution verified.

Original task/evaluator JSON, precondition project, solved project and full numerical CSV are included. No source evaluator or solver was executed locally. Independent replay, mesh convergence and physical-validity review remain pending.

Basic stress-test sample



Inspection render of original native FreeCAD Body BREP. Basic specimen, retained separately from complex cases. This is not an original dimensioned drawing.

Original native project / CSV	Verified content
CAD	1 solid; 14 faces; valid BREP; editable FCStd retained
Published load	2e+07 N
Result rows	805
Maximum displacement	811.423 mm
Peak von Mises stress	2.71792e+06 MPa

Saved FCStd includes displacement vectors, von Mises arrays, mesh and VTU results. All CSV values are finite. CSV displacement in m agrees with native mm; CSV stress in Pa agrees with native MPa. Source evaluator extrema match the CSV with its stated 5% tolerance.

20 MN force, AlZn4-5Mg1F35, Gmsh mesh max 5 mm/min 0.5 mm, CalculiX. About 811 mm displacement and 2.72 TPa stress make this unsuitable as validated elastic engineering ground truth. Retained as a basic workflow fixture; excluded from complex task counts.

Original task/evaluator JSON, precondition project, solved project and full numerical CSV are included. No source evaluator or solver was executed locally. Independent replay, mesh convergence and physical-validity review remain pending.